

HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE

SOFTWARE ENGINEER – HCL Technologies | Chennai, India

09/2021 – 01/2025 (3.5 Years)

- Designed and delivered **scalable data integration solutions** using **AWS EMR, S3, AWS Glue, and Lambda** to extract, transform, and distribute data across disparate enterprise systems, reducing data pipeline latency by **40%** and enabling real-time business intelligence for cross-functional stakeholders.
- Built and optimized **PySpark data pipelines** processing **500GB+ monthly datasets** across **Delta Tables, Parquet, and Avro file formats**, implementing **Apache Airflow** for orchestration and achieving **99.5% pipeline reliability** with automated error remediation.
- Implemented **Infrastructure as Code (IaC)** using **Terraform and CloudFormation** for AWS data warehousing infrastructure; automated deployment pipelines using **Jenkins, Git, Artifactory, YAML, and Maven**, reducing deployment time by **60%** and eliminating manual provisioning errors.
- Managed **data warehousing and data modeling** across **Databricks, Amazon Athena, RDS, and DynamoDB** to support complex analytical queries; designed dimensional schemas and optimized data lake structures for **sub-second query performance on petabyte-scale datasets**.
- Integrated heterogeneous data sources including **Oracle, PostgreSQL, MySQL, and NoSQL data stores** using **Java open-source standards and REST API frameworks**, building reusable connectors that reduced integration delivery time by **35%** across 12+ downstream product teams.
- Collaborated with cross-functional teams (Product Management, Integration Engineering, Quality Engineering, System Admin) using agile methodologies to gather requirements, troubleshoot complex data issues, and deliver high-performance distributed services; leveraged **GitHub Copilot** for accelerated development and testing cycles.

SKILLS

Data Integration & Big Data Technologies – AWS EMR, S3, AWS Glue, Lambda, Apache Airflow, PySpark, Scala, Spark engineering, data pipeline optimization.

Data Warehousing & Modeling – Databricks, Amazon Athena, RDS, DynamoDB, Delta Tables, Parquet, ORC, Avro, Sequence files, dimensional modeling, data lake architecture.

AWS Data Platforms & IaC – EC2, S3, Lambda, CloudWatch, Terraform, CloudFormation, Infrastructure as Code, cloud deployment patterns.

CI/CD & DevOps – Jenkins pipelines, Git, Artifactory, YAML, Maven, automated deployment, build automation for cloud environments.

Databases & Querying – Oracle, PostgreSQL, MySQL, SQL, NoSQL, Athena, RDS, complex data analysis, ad-hoc query optimization.

Programming & API Development – Java, Spring Boot, Python, REST APIs, open-source standards, distributed systems design, microservices integration.

Data Visualization & BI – Power BI, Tableau, DAX, Power Query, data modeling, business intelligence dashboards, analytics reporting.

EDUCATION AND TRAINING

PROJECTS

Enterprise Data Lake & Real-Time Analytics Platform (HCL Technologies)

- Architected and built a **petabyte-scale data lake** using **AWS S3, EMR, and Databricks** to consolidate workforce analytics data from **15+ heterogeneous sources** (Oracle ERP, PostgreSQL operational databases, DynamoDB event stores, and REST API integrations), reducing data acquisition time from **8 hours to 15 minutes**.
- Engineered **Apache Airflow orchestration workflows** with **PySpark transformations** to process **2TB daily incremental data** across **Delta Tables and Parquet formats**, implementing dynamic partition pruning and columnar compression to achieve **sub-second query latency** on analytical queries.
- Deployed infrastructure using **Terraform and CloudFormation** with **Jenkins CI/CD pipelines, Git version control, and Maven build automation**, enabling zero-downtime deployments and reducing infrastructure provisioning from **3 days to 2 hours**; integrated **GitHub Copilot** for code generation and testing acceleration.
- Designed star-schema **data warehouse models** in **Databricks and Amazon Athena** with optimized fact and dimension tables; engineered **AWS Glue jobs** for complex ETL logic and data quality validation, achieving **99.7% data accuracy** across reconciliation workflows.
- Integrated disparate data sources using **Java open-source frameworks and REST API standards**, building reusable **AWS Lambda-based connectors** for Oracle, PostgreSQL, MySQL, and DynamoDB; established **data governance frameworks** that reduced onboarding time for new data sources by **50%**.

SWIFT Payments Data Integration & Operational Analytics Dashboard (HCL Technologies)

- Managed data integration for a **SWIFT messaging and payments-operations platform**, extracting and transforming **ISO 20022 message formats (MT/MX standards)** from production payment flows; automated reconciliation workflows across **PostgreSQL, Oracle, and MySQL** to validate **10,000+ daily cross-border and SEPA transactions**.
- Built **Power BI dashboards** with **DAX measures** to visualize payment latency, transaction failure rates, and SLA compliance; delivered **real-time operational alerting** to banking operations teams, reducing mean-time-to-incident-resolution by **45%** and achieving **99.9% payment-processing uptime**.
- Collaborated with banking-domain business analysts and regulatory teams to ensure **GDPR compliance, audit trail integrity, and PSD2 regulatory requirements** were embedded in data pipelines; coordinated UAT cycles and post-implementation triage across interconnected enterprise systems.

ACHIEVEMENTS & CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California, Davis** (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)